

PROJECT DOCUMENTATION

Script-Controlled ACL – Restrict Record Access Based on Field Value

PHASE 4 – FUTURE ENHANCEMENT & OPTIMIZATION

Date	30 September 2026
Team ID	SWTID-2026-8983
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	1 Mark

Phase 4 Objective

This phase focuses on extending the implemented ACL solution, improving scalability, monitoring access activity, and optimizing the scripted access-control logic for future requirements.

Future Enhancements

S.No	Enhancement	Proposed Action	Expected Impact
1	Extend ACL coverage	Apply the access-control approach to additional tables and fields.	Broader record protection
2	Role + field validation	Combine user roles with field-value checks where required.	More flexible access control
3	Access monitoring	Maintain audit/reporting for restricted access attempts.	Better traceability
4	Performance optimization	Review and optimize script conditions for larger datasets.	Improved performance
5	Security review	Periodically review ACL rules and permissions.	Stronger security

Scalability Plan

Aspect	Current State	Future Plan
User Load	Suitable for current project users	Scale for larger user groups
Data Volume	Works with project-level records	Optimize queries and conditions for larger datasets
Performance	Scripted ACL validation	Minimize unnecessary processing
Security	Field-value based restriction	Add layered role and permission checks

Phase 4 Roadmap

Activity	Timeline	Status
Review current ACL implementation	Week 1	Planned
Extend access rules	Week 2	Planned
Performance and security testing	Week 3	Planned
Final documentation and deployment review	Week 4	Planned

Final Outcome

A scalable and maintainable Script-Controlled ACL solution with improved access control, monitoring, performance, and future extensibility.